Listing 4.14 Defining `email` as a `keyword` type for a `faculty` document

```
PUT faculty
{
  "mappings": {
    "properties": {
      "email": {
        "type": "keyword"
      }
    }
  }
}
```

Defines the email property

Declares email as a keyword type

We can also declare numeric values as keywords: for example, `credit_card_number` may be declared as a `keyword` for efficient access rather than as a numeric type such as `long`. There's no way we can build range queries on such data. The rule of thumb is that if numerical fields are *not* used in range queries, declaring them as `keyword` types is advised as it helps with faster retrieval.

> **NOTE** Sample code demonstrating the `keyword` data type is available with the book's files.

THE CONSTANT_KEYWORD TYPE

When the corpus of documents is expected to have the same value, regardless of number, the `constant_keyword` type comes in handy. Let's say the United Kingdom is carrying out a census in 2031, and for obvious reasons, the `country` field of each citizen's census document will be `"United Kingdom"` by default. There is no need to send the `country` field for each document when it is indexed into the `census` index. The mapping schema defines an index (`census`) with a field called `country` of type `constant_keyword`.

Listing 4.15 `census` index with `constant_keyword`

```
PUT census
{
  "mappings": {
    "properties": {
      "country":{
        "type": "constant_keyword",
        "value":"United Kingdom"
      }
    }
  }
}
```

Note that we set the default value for this field to `"United Kingdom"` when declaring the mapping definition. Now we index a document for John Doe, with just his name (no `country` field):